Gayatri Vidya Parishad College of Engineering Autonomous

Electronics and Communication Engineering | Course Code: EC202

Analog Electronic Circuits - Semester Exam Question Bank & Worked Soluti

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Syllabus Focus: EC202 High-Yield Semester Exam Question Bank & Model Solutions

Summary: Comprehensive question bank with previous semester university examination questions, m

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Small-Signal BJT & MOSFET Amplifiers

" Syllabus Scope: Hybrid-pi model, Common Emitter, Common Collector, High-frequency response, Miller effe

Unit 2: Feedback Amplifier Topologies

" Syllabus Scope: Voltage-series, Voltage-shunt, Current-series, Current-shunt feedback, Desensitivity fa

EC202: Analog Electronic Circuits - Advanced Topics & Assessment

Academic Track: Electronics and Communication Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Sinusoidal Oscillators

" Barkhausen criterion, RC phase-shift oscillator, Wien bridge oscillator, Hartley, Colpitts, and Quartz cryst

Unit 4: Operational Amplifiers & Comparators

" Differential amplifier, CMRR, Slew rate, Inverting and non-inverting configurations, Schmitt trigger, Precis

Unit 5: Active Filters & Power Amplifiers

" First and second order Sallen-Key low-pass and high-pass Butterworth filters, Class A, Class B push-pull, Cl

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the voltage gain, input impedance, and output impedance of a CE amplifier using the low-frequen
2. State Barkhausen criterion and derive the frequency of oscillation and minimum amplifier gain for a Wi
3. Design a second-order low-pass Sallen-Key active filter with a cutoff frequency of 5 kHz and Butterwor

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Adel S. Sedra, Kenneth C. Smith - Microelectronic Circuits Oxford University Press 7th Ed
* Jacob Millman, Christos C. Halkias - Integrated Electronics McGraw-Hill

Authorized NoteVault Educational Document " Verified for Registered University Students